ProveIt research report

Smaller explicit Diophantine forms of ZFC consistency

Twenty-four integers, tighter twelve-variable norms, and smaller materialized coefficients

AI-assisted mathematical synthesis
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Abstract

Combining rational Pell saturation, separating-root norms, signed-domain arguments and a concrete padding controller gives an explicit polynomial in 24 unrestricted integers of degree 1936 with a 167-operation circuit. ZFC proves that its lack of integer solutions is equivalent to Con(ZFC). Its three-equation version has degree 1920 and uses 162 operations. Three twelve-positive-variable alternatives have degree/operation pairs $(1069, 123)$, $(1073, 121)$ and $(1081, 120)$. An exact refactoring reduces the existing degree-1353 twelve-variable equation to 113 operations. Retaining the 943-tile language while replacing five-power rounding reduces the fixed base to $B = 2^{6364399}$; all four resulting integers are materialized. The report supplies explicit polynomial definitions, ordinary mathematical proofs, arithmetic circuits and reproducible finite audits. It does not claim a global minimum or a new kernel-checked end-to-end formalization.

Source snapshot: `ae3007ca4467011e0e051c84acb4a1dd13971b0e`.
The archive preserves source provenance and separates new proofs from inherited compiler and proof-calibration results.

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1 Results and the precise logical claim

There is no single useful meaning of "simplest" for these equations. Number of unknowns, quantified domain, total polynomial degree, arithmetic circuit length, and coefficient height are different resources. This report gives several improvements over the supplied collection, while preserving its calibrated blank-machine construction. The most substantial new integer-domain result is a polynomial in **24 unrestricted integers**, of exact total degree **1936**, with a **167-operation** evaluator. Its three-equation version has maximum degree 1920 and uses 162 operations.

For strictly positive integer unknowns, the report gives three new twelve-variable equations with degree/operation pairs

$$(1069, 123), \qquad (1073, 121), \qquad (1081, 120).$$

An exact refactoring of the existing degree-1353 twelve-variable polynomial also reduces its evaluator from 116 to 113 operations. A separate coefficient improvement reduces the radix exponent from 9,765,625 to 6,364,399, and supplies all four resulting fixed integers as actual binary payloads. These are complementary improvements, not a claim of a globally minimal Diophantine representation.

1.1 Comparison with the supplied snapshot

All comparisons refer to the repository snapshot `ae3007ca4467011e0e051c84acb4a1dd13971b0e`, rather than a moving `main`. The collection's own reconciliation identifies packages 16–23 as the current wave and records the historical alternatives [1]. Table 1 compares like domains and the particular resource being improved.

Single equation	Unknowns	Degree	Operations
Previous integer, least count and degree (19)	$26\mathbb{Z}$	2465	192
Previous integer, fewest ops at that count (21)	$26\mathbb{Z}$	2657	168
New signed separating norm	**$24\mathbb{Z}$**	**1936**	**167**
Previous positive, least degree at 12 (20)	12	1145	123
New tightest separating weight	12	1069	123
New intermediate weight	12	1073	121
New simpler weight	12	1081	120
Previous positive, fewest ops at 12 (21)	12	1353	116
Same polynomial, refactored here	12	1353	113
Previous monic merger (22)	13	319	91
Same polynomial, refactored here	13	319	89

Table 1: Positive integers unless marked $\mathbb{Z}$. The integer comparison is with the former minimum-variable family, not with every larger-variable equation in the collection. Source sizes: [3, 4, 5, 6].

The new 24-integer equation improves both recorded minima of the former 26-integer family: its degree is below 2465 and its operation count is below 168. Larger-variable integer equations still have much lower degrees; for example package 23 supplies 27 integers at degree 172 [7]. A lower degree is not being claimed across that tradeoff.

Two existing constructions also retain their particular advantages. Package 18 gives nine natural-number unknowns at degree $161844 \cdot 5^{33} + 484$, with 20,531 operations; translating every unknown

by one gives nine positive unknowns at the same degree [2]. The 15-positive-variable polynomial P_{67} remains a 67-operation equation, of degree 155. The retained-variable system from package 22 has 16 positive unknowns, seven equations of maximum degree 24, and 59 operations [9, 14]. Both of these latter circuits are included with the new complete coefficient tuple. Their algebra and operation counts are unchanged. Thus this report does not claim fewer than nine positive unknowns, fewer than 67 operations for a single equation, or fewer than 59 operations for a system.

1.2 The theorem is an equivalence inside ZFC

Fix the usual arithmetization of finite proofs in the chosen presentation of ZFC, and let

$$\mathrm{Con}(\mathrm{ZFC}) \equiv \neg\exists p\ \mathrm{Prf}_{\mathrm{ZFC}}(p, \ulcorner\bot\urcorner).$$

Here $\mathrm{Prf}_{\mathrm{ZFC}}$ is the finite syntactic proof predicate. The repository's proof predicate and any ordinary effective presentation are related by the standard proof translations formalizable in ZFC; no semantic soundness assumption is used in the equivalence.

Let $\boldsymbol{\kappa} = (\beta, K_0, M_C, M_F)$ be the four particular integers defined in Section 5. For the 24-variable polynomial P_{24} explicitly defined in Section 3, the claim is

$$\boxed{\mathrm{ZFC} \ \vdash\ \left(\neg\exists \boldsymbol{x} \in \mathbb{Z}^{24}\ P_{24}(\boldsymbol{x}; \boldsymbol{\kappa}) = 0\right) \ \longleftrightarrow\ \mathrm{Con}(\mathrm{ZFC}).} \tag{1}$$

The analogous statements hold over $\mathbb{Z}_{>0}^{12}$ for each displayed twelve-variable polynomial and over the corresponding domain for every listed system. Integer quantifiers here are an arithmetical abbreviation inside ZFC. The coefficients are fixed numerals; none is an extra unknown, an unspecified oracle, or a value defined by waiting for the machine to halt.

The construction proves the stronger positive equivalence

$$\exists \boldsymbol{x}\ P(\boldsymbol{x}) = 0 \quad\longleftrightarrow\quad \mathrm{BlankHalts}(K215) \quad\longleftrightarrow\quad \exists p\ \mathrm{Prf}_{\mathrm{ZFC}}(p, \ulcorner\bot\urcorner). \tag{2}$$

Negating this equivalence gives (1). In particular, one does not assume Con(ZFC) while proving the reduction, and this report does not prove the absence of solutions. Under the usual hypotheses of the second incompleteness theorem, if ZFC is consistent it cannot itself prove that absence. Bare consistency alone is not used to claim that ZFC cannot prove the opposite existential sentence; that additional assertion would require a stronger soundness hypothesis.

1.3 What is inherited, and what is proved here

The last equivalence in (2) is the calibrated K215 endpoint. The supplied Lean source names it `blank_halts_iff_zfc_refutation`, followed by `nonhalting_iff_zfc_consistency` [8]. A byte-identical copy and the machine table are included. This delivery does not rebuild that Lean development or represent the new number-theoretic reductions as kernel-checked theorems.

The inherited native-window compiler represents accepting blank-machine tableaux by Pell equations and fixed masks [9, 10, 12]. Its soundness applies to arbitrary solutions; its completeness constructs canonical witnesses from every finite accepting computation. The signed-domain and odd-scale results specify which coordinates need positivity and how square and odd substitutions preserve completeness [10, 11]. The source proof files needed for these interfaces accompany the report.

The new proofs supply three separate bridges. First, rational Pell saturation recovers an integer auxiliary root after root separation, allowing a smaller multiplier than the previous separating

construction. Tighter weights then cancel the top homogeneous terms in four of the sixteen linear conjugate factors. Second, a signed separation argument, including the $w = 0$ and $T = 0$ boundary cases, allows only six coordinates to retain positivity guards; the resulting lift uses 24 integers. Third, a concrete Boolean padding controller replaces the old five-power rounding of the coefficient layout. All three are ordinary mathematical arguments using finite arithmetic, elementary algebraic number theory, Pell identities and finite computation, hence formalizable in ZFC.

1.4 Conventions that keep the size claims meaningful

The degree is the total degree of the actual polynomial after all domain substitutions and after fixing the four coefficients. The maximum residual degree is reported for a system. A circuit charges every binary addition, subtraction and multiplication, including multiplication by a fixed numeral; using a previously computed value and comparing two values in a system are free. Squaring is one multiplication. The coefficient numerals are input constants, so their bit lengths are reported separately from circuit length.

Named intermediate expressions are polynomial abbreviations, not additional existential variables. Products over formal radicals below are short descriptions of integer polynomials. The delivered circuits evaluate them using only $+, -, \times$, and the report also gives explicit division-free formulas. No root extraction or resultant is an uncharged primitive. Positive-domain results are always labeled as such; a twelve-positive-variable equation is not asserted to have the same soundness over twelve unrestricted integers.

2 Twelve positive unknowns: a tighter separating norm

The construction combines the separating-root encoding of package 23 with rational Pell saturation from packages 19–22. Saturation removes the extra factor H in package 23's strong multiplier. A sharper separating weight then deliberately cancels leading terms in four conjugate factors. The result improves package 20's degree-1145, 123-operation equation [3, 4, 5, 7]. All statements in this section use the entire fixed coefficient tuple specified elsewhere in this report.

2.1 An explicit polynomial

The twelve supplied unknowns are

$$J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, t \in \mathbb{Z}_{>0}.$$

All other symbols are polynomial abbreviations. The fixed ports are β, K_0, M_C, M_F, with $\beta = B - 1$ and M_F the shifted high mask. Define

$$\begin{aligned}
&q = \beta J + 1, && C = q - F - \alpha, && X = wq, && Y = sq^3, && E = XY,\\
&k = \eta + \zeta, && c = kY + \eta, && a = E + Y, && H = 4a + 3, && \Delta = a^2 + H,\\
&\ell = EkY, && A = 1 + \ell(\ell + k), && G = 1 + \Delta c^2, && b_0 = X + ac.
\end{aligned}$$

The transport and index expressions are

$$\begin{aligned}
R &= ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J,\\
N_k &= k - hE - R,\\
N_t &= (K_0 + w)C + q - F - z(q-1).
\end{aligned}$$

The normalized strong and auxiliary expressions are

$$I = ic^2, \qquad S = \Delta I, \qquad Q = S^2, \qquad D = 1 + IS, \qquad d = Q - 1,$$
$$v = cT, \qquad b = c + RD, \qquad e = Q(Dv^2 + b^2) - 1, \qquad B_1 = 2Qvb.$$

In particular $\Delta = (a+2)^2 - 1$ and $D = 1 + \Delta I^2$. Choose one of the following three weights:

$$M_{1081} = 1 + 8I(T + 2REic), \tag{3}$$
$$M_{1073} = I\big(2T + R[E + 3(X+Y)]ic\big), \tag{4}$$
$$M_{1069} = I\big(2T + R[E + 2(X+Y) + sq^2]ic\big). \tag{5}$$

Set

$$m = HM, \qquad N = 1 + 2m, \qquad W = Ht + Nb_0.$$

For formal roots $\tau^2 = A$, $g^2 = G$, put

$$\mathcal{F}(\tau, g) = \big[d(W - m\tau - Ng)^2 - H^2 e\big]^2 - H^4 B_1^2 D, \tag{6}$$
$$\Omega = \prod_{\tau = \pm\sqrt{A},\, g = \pm\sqrt{G}} \mathcal{F}(\tau, g). \tag{7}$$

The product is an integer polynomial, without additional unknowns. The requested single equation is

$$\boxed{P = N_k N_t (1 + \Omega) - 1 = 0.} \tag{8}$$

The corresponding three-equation system is

$$N_k = 1, \qquad N_t = 1, \qquad \Omega = 0. \tag{9}$$

For clarity, no algebraic-number or resultant operation is needed to compute Ω. The following scalar circuit uses only integer arithmetic:

$$r_0 = W^2 + N^2 G, \qquad \Theta = 4W^2 N^2 G, \qquad a_0 = m^2 A,$$
$$C_0 = d(r_0 + a_0) - H^2 e, \qquad k_f = 4d^2 a_0, \qquad K_c = k_f r_0,$$
$$Z_0 = H^4 B_1^2 D, \qquad x_0 = C_0^2 + \Theta d^2 - K_c - Z_0, \qquad x_1 = 2C_0 d - k_f,$$
$$n_0 = x_0^2 + \Theta x_1^2 - 4Z_0 K_c, \qquad n_1 = 2x_0 x_1 - 4Z_0 k_f,$$
$$\Omega = n_0^2 - \Theta n_1^2.$$

To verify this identity, write $\theta = 2WNg$, so $\theta^2 = \Theta$. After eliminating τ, the conjugate factor is $(C^2 - K - Z_0)^2 - 4KZ_0$, where $C = C_0 + d\theta$ and $K = K_c + k_f\theta$. Its even and odd parts are n_0, n_1, respectively. The system compares $n_0^2 = \Theta n_1^2$ directly, saving the final subtraction. Shared intermediate values and the factorization of M are charged exactly as in the attached JSON circuits.

Theorem 2.1. *For each weight in* (3)–(5)*, the positive integer solvability of* (8) *or* (9) *is equivalent to blank halting of the calibrated K215 machine. Consequently ZFC proves that absence of a positive integer solution is equivalent to* Con(ZFC)*. The exact sizes are as follows; every binary addition, subtraction, or multiplication, including one by a fixed numeral, is charged.*

Weight	System degree	System operations	Equation degree	Operations
M_{1081}	1072	115	1081	120
M_{1073}	1064	116	1073	121
M_{1069}	1060	118	1069	123

The sum-of-squares equation $(N_k - 1)^2 + (N_t - 1)^2 + \Omega^2 = 0$ has degrees $2144, 2128, 2120$ and operation counts $123, 124, 126$, respectively. The theorem concerns ordinary mathematics formalizable in ZFC. It does not assert a new kernel-checked end-to-end reduction or prove consistency.

2.2 The integrality lemma that removes the extra multiplier

Lemma 2.2 (Rational Pell saturation)**.** *Let $s_0 \geq 2$ and V be integers. If $y \in \mathbb{Q}$ and*

$$s_0^2 V^2 - (s_0^2 - 1)y^2 = 1,$$

then $y \in \mathbb{Z}$.

Proof. Write $y = u/v_0$ in lowest terms with $v_0 > 0$. Its equation implies $v_0^2 \mid s_0^2 - 1$. The integer $D_0 = (s_0^2 - 1)/v_0^2$ is positive and nonsquare, since otherwise $s_0^2 - 1$ would be a square. The pairs (s_0, v_0) and $(s_0|V|, |u|)$ solve $x^2 - D_0 z^2 = 1$ with positive ordinates; $u = 0$ or $V = 0$ is impossible. Write them as (χ_j, ψ_j) and (χ_l, ψ_l) in the powers of the fundamental positive integer-coordinate Pell unit, where $j, l \geq 1$.

We have $\chi_j \mid \chi_l$. The addition identities imply

$$\chi_{r+2j} \equiv -\chi_r \pmod{\chi_j}, \qquad \chi_{2j-r} \equiv -\chi_r \pmod{\chi_j}.$$

Reduce l modulo $2j$, then reflect to $0 \leq r \leq j$. Strict growth gives $0 < \chi_r < \chi_j$ for $r < j$, so the divisible case requires $r = j$. Thus l is an odd multiple of j. The Pell multiplication formula gives $\psi_j \mid \psi_l$, hence $v_0 \mid u$. Their coprimality forces $v_0 = 1$. □

This is the special feature of the auxiliary norm. Rationality of a root of a general nonmonic quadratic would not imply integrality. Packages 19–22 establish equivalent saturation lemmas [3, 4, 5]; applying the lemma before recovering the main congruence removes package 23's extra H without changing any canonical compiler witness.

2.3 Separation bounds before recovering roots

First, Θ is divisible by 4, so Ω is a square modulo 4. At a zero of (8), its three integer factors are units. The last cannot be -1: that would require $\Omega = -2$, which is not a square modulo 4. Therefore

$$\Omega = 0, \qquad N_k = N_t = \epsilon \in \{1, -1\}.$$

The system has $\epsilon = 1$. The inherited weak transport bootstrap is valid for both signs and gives, before any eliminated root is recovered,

$$\begin{gathered} C \geq 0, \quad q \geq 16, \quad X \geq 16, \quad Y \geq q^3 \geq 4096, \\ 0 < R < q^4 - 2, \qquad E \geq q^4 > R + 2. \end{gathered} \tag{10}$$

These are coefficient-dependent statements from the authentic blank interface, not assertions for arbitrary substituted ports [10, 4, 5].

A vanished factor in (7) supplies coherent roots satisfying

$$\tau^2 = A, \qquad g^2 = G, \qquad f^2 = D,$$

$$dy^2 = e - B_1 f = Q(vf - b)^2 - 1, \tag{11}$$

$$t = y + M\tau + N\gamma, \qquad \gamma = (g - b_0)/H. \tag{12}$$

The roots here are initially algebraic. In particular, no integrality or sign of γ has been assumed.

For the first weight, all complex auxiliary branches satisfy

$$|y| < 2\big(v(a+2)I + b\big).$$

Indeed $\sqrt{D} < (a+2)I$ and $Q \geq 2$ give this bound directly from (11). Since $a+2 < 2E$, $c < 2kY$, $\sqrt{A} > \ell > Ec/2$, and $D < 4E^2I^2$,

$$\frac{|y|}{\sqrt{A}} < 8TI + \frac{4}{E} + 16REi^2c^3 < M_{1081}.$$

Here is the sharper estimate used for the other weights. Since $\Delta > R+2$ and $I^2 \geq c^4$,

$$Q - b = \Delta I^2(\Delta - R) - c - R > 0.$$

Thus $b \leq d$; also $\sqrt{D} < S < d$. Put $Z = v\sqrt{D} + b > 1$. For every auxiliary conjugate,

$$|y|^2 \leq \frac{QZ^2+1}{d}, \qquad \left[Z\left(1+\frac{1}{d}\right)\right]^2 - \frac{QZ^2+1}{d} = \frac{QZ^2-d}{d^2} > 0.$$

Consequently

$$|y| < Z + v + 1 < cT\big((a+2)I+1\big) + c + RD + 1. \tag{13}$$

Let $\rho = (1+1/Y)/E$. We have $c/\ell < \rho$, and $(a+3)\rho < 2$, because $(a+3)/E < 9/8$ and $1 + 1/Y < 17/16$. Furthermore

$$(a+2)^2/E = E + 2Y + 4 + Y/X + 4/X + 4/E < E + 3Y + 5,$$
$$(a+2)^2\rho < E + 3Y + X + 9 < E + 3(X+Y) - 2.$$

For $C_1 = E + 3(X+Y)$, dividing (13) by $\sqrt{A}$ therefore gives

$$\frac{|y|}{\sqrt{A}} < 2TI + R(C_1 - 2)i^2c^3 + 2 \leq 2TI + RC_1i^2c^3 = M_{1073}.$$

The last inequality uses $Ri^2c^3 \geq 1$.

For $C_2 = E + 2(X+Y) + sq^2$, use the exact identity

$$(a+2)^2\rho = E + 2Y + X + 6 + Y/X + 5/X + 4/Y + 8/E + 4/(EY).$$

Since $Y/X = sq^2/w \leq sq^2$ and $5/X + 4/Y + 8/E + 4/(EY) < 1$, the right side is less than $E + 2Y + X + sq^2 + 7 < C_2 - 2$. The identical argument proves $|y| < M_{1069}\sqrt{A}$.

Finally $G > A$: $c > kY$ and $\Delta - E^2 > 2EY$ imply

$$\Delta c^2 > k^2Y^2\Delta > k^2(E^2Y^2 + EY) = A - 1.$$

For each of the three weights, the two uniform separation inequalities are

$$|y| < M\sqrt{A}, \qquad \frac{N\sqrt{G}}{H} > 2M\sqrt{G} > 2M\sqrt{A}. \tag{14}$$

2.4 Rationality, integrality, and the compiler interface

Work in a finite normal splitting field over $\mathbb{Q}$ containing all roots in (11). If an automorphism changed the sign of g, it would change $N\gamma$ by magnitude $2N\sqrt{G}/H$. The change in $M\tau$ has magnitude at most $2M\sqrt{A}$, and the change in y has magnitude strictly less than $2M\sqrt{A}$, using the uniform bound for both conjugates. Their sum cannot cancel the former change by (14), contradicting the fixed rational value t in (12). Hence g is rational. With g fixed, a change in τ would similarly

require $2M\sqrt{A} < 2M\sqrt{A}$. Thus τ is rational. Equation (12) makes y rational, and $B_1 > 0$ makes $f = (e - dy^2)/B_1$ rational.

Since $g^2 = G$, $\tau^2 = A$, and $f^2 = D$ are integers, g, τ, f are integers. The integer $V = vf - b$ and $S = \Delta I \geq 2$ now satisfy the hypotheses of Lemma 2.2, so y is an integer. Both $H\gamma = g - b_0$ and $N\gamma = t - M\tau - y$ are integers, and $N = 1 + 2HM$ gives $\gcd(H, N) = 1$. Bezout's identity proves $\gamma \in \mathbb{Z}$.

The recovered equations are precisely the normalized first, main, strong, and auxiliary Pell equations. The inherited signed-coordinate theorem permits signed γ, τ, f, y, T, and conversion to its ordinary strong coordinate is $i_{\mathrm{ord}} = \Delta i$. That theorem restores the unit sign $\epsilon = 1$ and extracts a blank K215 halting computation [10, 4, 5]. In particular, the sign of the main root is not assumed in the reconstruction.

Conversely, take a canonical positive normalized blank compiler witness. For each chosen weight, set

$$t = y + M\tau + N\gamma > 0.$$

One factor of (7) vanishes and $N_k = N_t = 1$. No additional divisibility restriction on i and no adjustment of the auxiliary Pell period is required. This proves the existential equivalence in Theorem 2.1; composing with the calibrated K215 refutation equivalence and negating existential solvability gives its claimed ZFC theorem.

2.5 Uniform exact degrees and the deliberate cancellation

Scale the twelve supplied variables by a formal parameter λ and write a subscript 0 for the leading homogeneous part. The shared degrees are

expression	E, H	Y	c	R	Δ	I	Q, d	D	b	A, G	M, N
degree	6	4	5	4	12	11	46	34	38	22	27, 33.

Put

$$\mathcal{B} = R_0 E_0^2 i^2 c_0^4, \qquad \mathcal{D} = R_0 E_0 Y_0 i^2 c_0^4.$$

These have degrees 38 and 36. The two formal main-quotient roots solve

$$H^2\gamma^2 + 2Hb_0\gamma + (b_0^2 - G) = 0.$$

The coefficient degrees are $12, 17, 16$, with $(b_0^2 - G)_{16} = -H_0 c_0^2 \neq 0$. Thus the two roots have orders 5 and -1; the larger has leading term $-2\ell_0/H_0$. Also each auxiliary root has expansion

$$y = \varepsilon(b - vf) + O(\lambda^{-8}), \qquad \varepsilon \in \{1, -1\},$$

since $Q/(Q-1) = 1 + O(\lambda^{-46})$, while vf has order 23. These expansions can be taken in a Laurent-series extension of the rational function field; all leading radicands are squares there.

For M_{1081}, $M\sqrt{A}$ has leading term $16\mathcal{B}$ and $N\gamma$ on its large branch has leading term $-64\mathcal{B}$. The exact factorization is

$$\Omega = d^8 H^{16} \prod (t - y - M\tau - N\gamma), \tag{15}$$

with sixteen coherent branches, counting multiplicity. Their leading magnitudes are $15, 17$ four times each, and $47, 49, 79, 81$ twice each, in units of $\mathcal{B}$. No leading term vanishes. Hence

$$\deg \Omega = 8 \cdot 46 + 16 \cdot 6 + 16 \cdot 38 = 1072.$$

The leading coefficient multiplier is $15^4 17^4 (47 \cdot 49 \cdot 79 \cdot 81)^2$, which is nonzero.

For M_{1073}, the positive formal first root satisfies $\sqrt{A} - \ell = O(\lambda)$, and

$$M\sqrt{A} - b = Ri^2c^3(C_1\ell - \Delta c) + O(\lambda^{28}).$$

Here the exact polynomial identity

$$C_1\ell - \Delta c = (EY + 3EX - Y^2 - 4E - 4Y - 3)c - E(E + 3X + 3Y)\eta$$

has unique leading form $E_0Y_0c_0$ of degree 15. Consequently $M\sqrt{A} - b$ has leading form $\mathcal{D}$ of degree 36: its terms of degrees 38 and 37 cancel identically. Now $N\gamma$ on the large branch has leading term $-4\mathcal{B}$; on the small branch it has order 32. In (15), four factors have degree 36 and leading magnitude $\mathcal{D}$, four have degree 38 and magnitude $2\mathcal{B}$, and the remaining eight have leading multiples $2, 4, 4, 6$ of $\mathcal{B}$, each twice. The signed product is positive:

$$\Omega_{1064} = 589824\, d_0^8 H_0^{16} \mathcal{B}^{12} \mathcal{D}^4,$$
$$1064 = 8 \cdot 46 + 16 \cdot 6 + 12 \cdot 38 + 4 \cdot 36.$$

Writing $q_0 = \beta J$, $k_0 = \eta + \zeta$ gives the fully substituted form

$$\Omega_{1064} = 9 \cdot 2^{48} i^{48} k_0^{96} w^{76} s^{176} q_0^{652} (q_0 - F)^{16}. \tag{16}$$

It is nonzero for every authentic slice $\beta > 0$. The product's leading form multiplies this by

$$[-hwsq_0^4]\,[w(q_0 - F - \alpha) - zq_0].$$

For example, the coefficient of $J^{673} hzi^{48} \eta^{96} w^{77} s^{177}$ is $9 \cdot 2^{48} \beta^{673} > 0$. This proves exact degree 1073 uniformly, not merely for diagnostic numerical specializations.

Finally, for M_{1069} replace C_1 by C_2 above. Then

$$C_2\ell - \Delta c = (2EX + Esq^2 - Y^2 - 4E - 4Y - 3)c - EC_2\eta.$$

Its unique leading term is $E_0(sq_0^2)c_0$, of degree 14. Thus the four cancellation factors have degree 35, rather than 36. All other factors retain their previous degrees and leading multipliers. The norm has exact degree 1060, and its leading form is

$$\Omega_{1060} = 9 \cdot 2^{48} i^{48} k_0^{96} w^{76} s^{176} q_0^{648} (q_0 - F)^{16}.$$

The corresponding product coefficient at $J^{669} hzi^{48} \eta^{96} w^{77} s^{177}$ is $9 \cdot 2^{48} \beta^{669} > 0$, proving exact degree 1069. The degrees of N_k, N_t are $7, 2$ in every case. The sum-of-squares claims follow because Ω^2 is its unique highest-degree summand.

The independent finite checks compare the literal circuits against conjugate-product evaluators, verify all gates and ports are live, and compute full univariate specializations over two finite fields to detect the cancellations. They corroborate the formulas and operation counts; the preceding uniform arguments establish the mathematical equivalence and exact degrees. They do not construct an accepting K215 computation.

3 Twenty-four unrestricted integer unknowns

The preceding construction can be adapted to a substantially smaller integer domain. The current collection has equations in 26 unrestricted integers; the construction below has **24**, degree **1936**, and a division-free evaluator with **167** binary operations. Its three-equation version uses 162 operations and has maximum degree 1920. The proof combines the saturated root norm of package 19 with the odd-scale signed-domain theorem of package 23 [3, 7, 11]. The new work is the separation argument when the half-binomial scale, the main Pell parameter, and the separating coefficients can be negative.

3.1 Domain maps and the equation

For each of the six ports

$$J, F, \alpha, z, \eta, \zeta$$

use a disjoint copy of the polynomial

$$\mathrm{Pos}(x, y, u) = 1 + x^2 + y^2 + u(u+1).$$

Its values are positive, and it represents every positive integer: apply the three-square theorem to $4n-3$, whose representation has one odd square and two even squares. Supply six further unrestricted integers h, T, t, r_i, r_s, r_w, and put

$$i = 2r_i + 1, \qquad s = 2r_s + 1, \qquad w = r_w^2. \tag{17}$$

Thus the number of supplied integers is $6 \cdot 3 + 6 = 24$. The executable circuit names these last three inputs `ii`, `ss`, and `ww`, and names the merged coordinate t `r`.

Retain the common abbreviations $q, C, X, Y, E, k, c, a, H, \Delta, R, N_k, N_t$ of the positive construction. In particular,

$$\begin{aligned} X &= wq, & Y &= sq^3, & E &= XY, & k &= \eta + \zeta, \\ c &= kY + \eta, & a &= E + Y, & H &= 4a + 3, & \Delta &= a^2 + H. \end{aligned}$$

To make the norm definition explicit, set

$$\begin{aligned} L &= kXY^2, & A &= 1 + L(L+k), & G &= 1 + \Delta c^2, \\ P &= X + ac, & I &= ic^2, & D &= 1 + \Delta I^2, \\ Q &= \Delta^2 I^2, & d &= Q - 1, & v &= cT, \\ b &= c + RD, & e &= Q(Dv^2 + b^2) - 1, & B_1 &= 2Qvb. \end{aligned}$$

Here L is an arithmetic abbreviation, unrelated to a compiler cell length. All these symbols are computed expressions, not new unknowns.

Use the following signed separating weights:

$$M = 1 + 8(TI)^2 + 16Rai^2c^3 = 1 + 8(ic)I(T^2c + 2Ra), \tag{18}$$

$$N = 1 + 8HM, \qquad Z = Ht + NP. \tag{19}$$

The second expression for M is the factored form used by the circuit. For formal roots $u^2 = A$ and $g^2 = G$, define

$$F(u, g) = \left[d(Z - HMu - Ng)^2 - H^2 e\right]^2 - H^4 B_1^2 D, \tag{20}$$

$$\Omega = \prod_{u=\pm\sqrt{A}} \prod_{g=\pm\sqrt{G}} F(u, g). \tag{21}$$

The product is an ordinary polynomial with integer coefficients; its division-free circuit does not use square roots or a resultant operation. The requested single equation is

$$\boxed{N_k N_t (1 + \Omega) - 1 = 0.} \tag{22}$$

The corresponding system is $N_k = 1$, $N_t = 1$, $\Omega = 0$. All four fixed coefficient ports must come from one complete authentic tuple. The new tuple in Section 5 has the required interface; package 19's compact Wang943 tuple is a separate valid choice.

Theorem 3.1. *For the coherent fixed coefficients specified in this report, equation (22) has an integer zero if and only if the blank run of K215 halts. The same statement holds for its three-equation system. Consequently ZFC proves that the absence of integer zeros is equivalent to* Con(ZFC).

3.2 Bootstrap with signed scales

The optimized norm evaluator has the form $\Omega = U^2 - \Theta V^2$, where $\Theta = 4Z^2N^2G$. Thus Ω is a square modulo four. At a zero of (22), all three integer factors are units. The alternative $1+\Omega = -1$ would require a square to equal two modulo four. Therefore

$$\Omega = 0, \qquad N_k = N_t = \epsilon, \qquad \epsilon \in \{1, -1\}.$$

For the system these conclusions hold directly with $\epsilon = 1$.

The weak transport estimate needs only the six guarded positive ports:

$$q - F - z(q-1) = 1 - F - (z-1)(q-1) \le 0.$$

If $C \le -1$, then $N_t \le -(K_0 + w) \le -2$, a contradiction. Hence $C \ge 0$ and $q - F = C + \alpha > 0$. The authentic mask bounds give

$$q \ge 16, \qquad q(q^2-1) < R < q^4 - 2. \tag{23}$$

These estimates are valid before reconstructing any root and do not use $s > 0$ [10, 3].

Since s is odd, $|Y| \ge q^3$. Also $X \ge 0$, and a, c have the same sign as Y. Their absolute values satisfy $|a| \ge q^3$ and $|c| > k(|Y| - 1) > 1$. Consequently $|H| > 1$ and

$$\Delta = (a+2)^2 - 1 \ge (q^3 - 2)^2 - 1 > q^4 > R.$$

The multiplier i is nonzero, so

$$D > \Delta + |c|, \qquad 1 < b < Q, \qquad Q > D > 1. \tag{24}$$

For the upper bound on b, use $Q - b = (\Delta - R)D - \Delta - c > 0$. In particular b is positive even when c is negative. Finally, $Rac^3 > 0$, so $M > 1$, and

$$\left|\frac{N}{H}\right| > 7M, \qquad \gcd(N, H) = 1. \tag{25}$$

3.3 Recovering roots without assuming their signs

Because $A \ge 1$ and $G > 1$, a zero of (21) selects real roots u, g with $F(u, g) = 0$. Work in the totally real Galois field $\mathbb{Q}(u, g)$, allowing degrees one, two, or four, and put

$$y = \frac{Z - HMu - Ng}{H}.$$

When $B_1 \neq 0$, define $f = (e - dy^2)/B_1$. Then $f^2 = D$ and every actual conjugate satisfies $d(y')^2 = Q(vf' - b)^2 - 1$. When $B_1 = 0$, necessarily $v = 0$, and the same factor directly gives $d(y')^2 = Qb^2 - 1$. In both cases,

$$|y'| < 2\big(|v|\sqrt{D} + b\big), \qquad Z = HMu + Ng + Hy. \tag{26}$$

No division by T or B_1 is made in the exceptional branch.

First exclude $w = 0$. The inequalities $|a| \ge 4096$ and $|c| > 1$ give

$$\sqrt{G} > \frac{|a||c|}{2}, \qquad \sqrt{D} < 2|a||i|c^2, \qquad D < 4a^2i^2c^4.$$

Term by term in (18), these imply

$$\begin{aligned} M\sqrt{G} &> 2|c| + 4|a||iT||c|^3 + 8Ra^2i^2c^4 \\ &> 2(|v|\sqrt{D} + b). \end{aligned} \tag{27}$$

Here $i^2T^2c^2 \geq |iT|$, including $T = 0$, and $b \leq |c| + RD$. If $w = 0$, then $A = 1$ and u is already integral. An automorphism flipping g would contradict (25), (26), and (27). Thus g is rational and integral, and y is rational.

We use the rational auxiliary Pell saturation lemma: for integers $S_0 > 1, V$, a rational solution of

$$S_0^2V^2 - (S_0^2 - 1)y^2 = 1 \tag{28}$$

is integral. Its proof by ordinary Pell classification and divisibility of first coordinates does not assume $S_0^2 - 1$ squarefree [3]. When $B_1 \neq 0$, the rational f has integral square, so is integral, and the lemma applies with $S_0 = |\Delta I|$, $V = vf - b$. When $B_1 = 0$, apply it with $V = -b$. Thus y is integral in either case.

Set $\gamma = (g - P)/H$. The linear equation gives $t = Mu + N\gamma + y$. Hence $H\gamma$ and $N\gamma$ are integers; coprimality in (25) makes γ integral. At $X = 0$ the main norm would now imply

$$1 = (ac + H\gamma)^2 - \Delta c^2 = H(2ac\gamma + H\gamma^2 - c^2),$$

contradicting $|H| > 1$. Therefore $w > 0$ and $X \geq q \geq 16$.

For $w > 0$, we have $\sqrt{A} > L$, $|a + 2| < 2|E|$, and $|c| < 2k|Y|$. Dividing (26) by $\sqrt{A}$ and using $L = |E|k|Y|$ yields

$$\begin{aligned} \frac{|y'|}{\sqrt{A}} &< 8|TI| + \frac{4}{|E|} + 16R|E|i^2|c|^3 \\ &< 1 + 8(TI)^2 + 16R|a|i^2|c|^3 = M. \end{aligned} \tag{29}$$

The last step uses $|E| \geq q^4$, $|a| > |E|$, and the integer inequality $(TI)^2 \geq |TI|$.

For either sign of Y, the elementary estimates

$$\Delta > \tfrac{3}{4}a^2, \quad |c| > \tfrac{3}{4}k|Y|, \quad G > \tfrac{27}{64}L^2, \quad A < \tfrac{3}{2}L^2$$

give $\sqrt{G} > \sqrt{A}/2$. An automorphism flipping g in (26) would therefore imply $|N/H|\sqrt{G} < 2M\sqrt{A}$, whereas (25) gives $|N/H|\sqrt{G} > \frac{7}{2}M\sqrt{A}$. Thus g is rational. Once g is fixed, a flip of u contradicts the uniform bound (29). Hence u is rational as well. Both are integers, and the saturation and coprimality argument above makes y, γ integers. This uses only actual field automorphisms; square radicands or coincident quadratic extensions cause no exception.

If $T = 0$, the recovered integral y would satisfy

$$(Q - 1)(y^2 - b^2) = b^2 - 1 > 0.$$

Since $1 < b < Q$, one has $|y| \geq b + 1$ and $Q - 1 \geq b$. The left side is at least $b(2b + 1) > b^2 - 1$, a contradiction. Thus $T \neq 0$, and the recovered $f = (e - dy^2)/B_1$ is an integer.

3.4 Soundness and completeness of the signed domain

The roots just reconstructed satisfy exactly

$$\begin{aligned} u^2 &= 1 + L(L+k), \\ (X + ac + H\gamma)^2 - \Delta c^2 &= 1, \\ f^2 - \Delta i^2 c^4 &= 1, \\ Q[c(Tf-1) - Rf^2]^2 - (Q-1)y^2 &= 1, \qquad N_k = N_t = \epsilon. \end{aligned}$$

There are six positive guarded ports, odd s, i, nonzero square w, and unrestricted integer roots. These are precisely the hypotheses of the signed odd-scale theorem [11]. Its negative-s argument recovers the main Pell rank R and rules out both main-root signs by their congruences modulo H. Hence $s > 0$. The inherited signed compiler then forces $\epsilon = 1$ and recovers an accepting blank K215 tableau [10]. Our root recovery did not assume these conclusions.

Conversely, an accepting computation has canonical normalized witnesses with odd i and square w. The additional oddness of s follows from the native population identities. If $q = 2^j$, the native masks give

$$\operatorname{popcount}(R) = 3j + 2, \qquad v_2(Y) = \operatorname{popcount}(R) - 2 = 3j.$$

Since $Y = sq^3$, the scale s is odd. The required mask hypotheses are $M_C \equiv 2 \pmod 4$, $M_F^{\text{native}} \equiv 4 \pmod 8$, and $\operatorname{popcount}(M_C) + \operatorname{popcount}(M_F^{\text{native}}) = d_{\text{base}} = \log_2 B$; the coefficient adapter preserves them. The arithmetic port is still the *shifted* M_F.

Choose $t = y + Mu + N\gamma$. Then one norm factor vanishes and $N_k = N_t = 1$. The six positive coordinates admit their three-integer representations. This proves Theorem 3.1 after composition with the inherited K215/refutation calibration.

3.5 Exact degree and operation count

Assign weight two to $J, F, \alpha, z, \eta, \zeta, w$ and weight one to s, i, h, T, t. The principal degrees are

$$\begin{aligned} \deg(a, H, c, \Delta, L, A, G) &= (11, 11, 9, 22, 20, 40, 40), \\ \deg(R, I, D, Q, d, b, e, B_1) &= (8, 19, 60, 82, 82, 68, 218, 160), \\ \deg(M, N, Z, N_k, N_t) &= (48, 59, 79, 12, 4). \end{aligned}$$

Write a subscript 0 for the highest homogeneous part. Then $A_0 = G_0 = L_0^2$, $P_0 = L_0 = a_0 c_0$, and

$$M_0 L_0 = 16 b_0, \qquad N_0 = 8 H_0 M_0.$$

In the four leading sign choices, $Z - HMu - Ng$ has leading part $H_0 M_0 L_0$ times one of $-1, 1, 15, 17$. Thus the leading coefficients of the inner bracket in (20), after extracting $H_0^2 d_0 b_0^2$, are

$$255, \quad 255, \quad 57599, \quad 73983.$$

Their squares have degree 480, while the competing term $H^4 B_1^2 D$ has degree 424. Therefore

$$\Omega_0 = C_* H_0^{16} d_0^8 b_0^{16}, \qquad C_* = 255^4 (57599 \cdot 73983)^2 > 0. \tag{30}$$

Its degree is $16 \cdot 11 + 8 \cdot 82 + 16 \cdot 68 = 1920$. Multiplying $N_k N_t$ adds $12 + 4$, giving exact degree 1936. The squared-residual alternative has degree 3840.

For an explicit uniform certificate, in the actual circuit variable names the coefficient of

$$J_x^{1354} h\,\mathrm{ii}^{48} \eta_x^{192} \mathrm{ww}^{162} \mathrm{ss}^{177} z_x^2$$

is $2^{257} C_* \beta^{677}$. The exponent sum is 1936 and the coefficient is nonzero for every authentic $\beta > 0$. Thus the degree claim applies to the full fixed integers, independently of numerical specializations.

The optimized positive core takes 120 product operations; the signed weight takes two more. The domain maps initially cost 47 operations. Two additions disappear by absorbing the constant terms of F and z: if $F = 1 + F_*$, $z = 1 + z_*$, and $r = \beta J$, then

$$q - F = r - F_*, \qquad N_t = (K_0 + w)C - F_* - z_* r.$$

The final counts are therefore 167 operations for the product, 162 for the system, and 170 for the sum of squares. All supplied coordinates and all emitted gates are live. Exact conjugate-product checks, domain-map identity checks, and independent finite-field leading-coefficient checks corroborate the circuit. The universal equivalence rests on the proof above and the explicitly stated inherited dependencies, rather than on those finite diagnostics.

4 Division-free circuits and arithmetic savings

All displayed auxiliary symbols in this section denote polynomial abbreviations. They introduce neither existentially quantified coordinates nor operations outside addition, subtraction, and multiplication. The delivered JSON files specify their evaluation order and every operand. The accounting convention charges each binary arithmetic operation, including multiplication by a fixed integer; reusing an earlier value and comparing two computed expressions are free. This is the convention used by the repository comparison circuits [5, 6]. All polynomial-identity comparisons hold the four coefficient ports equal. Replacing the whole tuple is a separate semantic step proved in Section 5.

4.1 A shared evaluator for the four-root construction

Use the notation of the root-recovery construction, and write W for its supplied merged expression, N for the weight of the main root, and $m = HM$ for the weight of the first Pell root. Thus the relevant eight-conjugate product is

$$\mathcal{R} = \prod_{\sigma,\epsilon,\delta \in \{-1,1\}} \left[d(W + \sigma N\sqrt{G} + \epsilon m\sqrt{A})^2 - H^2 e + \delta H^2 B_{\mathrm{aux}} \sqrt{D} \right]. \tag{31}$$

Here $B_{\mathrm{aux}} = 2Qvb$. Define $P_a = QDv^2$ and $U_a = Qb^2$, so that $B_{\mathrm{aux}}^2 D = 4P_a U_a$. Formula (31) explains the elimination; it is not the circuit definition and does not require a square-root operation.

The following completely integral evaluator computes the same polynomial:

$$\begin{aligned}
V_0 &= W^2 + N^2 G, & J_\theta &= 4W^2(N^2 G), & a_0 &= m^2 A,\\
C_0 &= d(V_0 + a_0) - H^2 e, & K_* &= 4d^2 a_0, & U_0 &= K_* V_0,\\
Z_0 &= 4H^4 P_a U_a,\\
x &= C_0^2 + J_\theta d^2 - U_0 - Z_0, & y &= 2C_0 d - K_*,\\
n_0 &= x^2 + J_\theta y^2 - 4Z_0 U_0, & n_1 &= 2xy - 4Z_0 K_*,\\
\mathcal{R} &= n_0^2 - J_\theta n_1^2.
\end{aligned}$$

The three-equation system compares

$$N_k = 1, \qquad N_t = 1, \qquad n_0^2 = J_\theta n_1^2.$$

The corresponding single equation is

$$N_k N_t (1 + \mathcal{R}) - 1 = 0.$$

Since $4 \mid J_\theta$, the integer $\mathcal{R}$ is a square modulo 4. Consequently $1 + \mathcal{R}$ cannot equal -1. This preserves the modular guard needed in the product equation; the remaining outer signs are handled by the root-recovery proof.

To verify the evaluator, work temporarily in a quadratic algebra and put $\theta = 2WN\sqrt{G}$, so that $\theta^2 = J_\theta$ and $(W + N\sqrt{G})^2 = V_0 + \theta$. Taking the norm in the first-root direction uses the exact identity

$$\big((C-u)^2 - Z\big)\big((C+u)^2 - Z\big) = (C^2 - u^2 - Z)^2 - 4u^2 Z. \tag{32}$$

After the first two eliminations, the remaining quadratic-algebra element is $n_0 + n_1\theta$. Its norm is the last line of the evaluator. This is an identity in a free integer-polynomial ring. In particular no division by W, N, or $2WN$ is performed, and the identity remains valid when one of these quantities is zero.

4.2 Factoring the common odd coefficient

The earlier paired evaluator explicitly constructs $V_1 = 2WN$, then dV_1, K_*V_1, and their products and squares. Every odd component has this same factor V_1. Retaining only its square $GV_1^2 = 4W^2(N^2G)$ avoids three arithmetic operations in the complete polynomial evaluator. Both W^2 and N^2G are already needed for V_0. The system saves one further subtraction by comparing its two final norm terms directly.

Applied to the unchanged nested-norm polynomial of package 21, this gives

$$(12 \text{ positive variables},\ 1 \text{ equation},\ 1353,\ 113)$$

in the order (unknowns, equations, degree, operations), improving its 116-operation evaluator. Its three-equation system uses 108 operations rather than 112, with the same maximum degree 1344. The polynomial itself, its zero set, and the proof assumptions are unchanged [5].

The same idea applies to the monic-resultant circuits of package 22. There $r_1 = 2W$ occurs in every odd component, while Br_1^2 already occurs in the even component. In its twelve-variable evaluator, write

$$v_I = r_1 t, \qquad h_I = 2r_1(v_R t - \lambda u_R).$$

One may compute $Bv_I^2 = (Br_1^2)t^2$ and $Bh_I^2 = 4(Br_1^2)(v_R t - \lambda u_R)^2$ directly. Moreover $Br_1^2 = 4BW^2$ needs two gates because W^2 is already available, whereas forming r_1, squaring it, and multiplying by B uses three. The complete saving is three operations: 117 becomes 114 for the polynomial, and 112 becomes 109 for its system. The thirteen-variable intermediate has $h_I = r_1 t$ and saves two operations: its degree-319 polynomial uses 89 operations, and its four-equation system of maximum degree 288 uses 81. Again these are exact refactorings of the original polynomials [6].

4.3 Weights and the unrestricted-integer lift

The new positive constructions use the same shared evaluator with different separating weights. For the degree-1081 version, the identity

$$1 + 8TI + 16REi^2c^3 = 1 + 8I(T + 2REic), \qquad I = ic^2,$$

saves one more operation. The tighter degree-1073 and degree-1069 weights are evaluated in their already factored form

$$M = I(2T + R\mathcal{C}\,ic),$$

with $\mathcal{C} = E + 3(X + Y)$ and $\mathcal{C} = E + 2(X + Y) + sq^2$, respectively. Their improvement in degree comes from the cancellation proved in the weight analysis, rather than from the generic degree bound of the evaluation graph.

For the signed construction use

$$M = 1 + 8(TI)^2 + 16Rai^2c^3 = 1 + 8(ic)I(T^2c + 2Ra), \qquad N = 1 + 8HM.$$

The second expression is a polynomial identity for arbitrary integers, including negative i and c. The separate signed-domain proof supplies the positivity properties needed in the root argument.

There is a further saving when the six guarded coordinates are replaced by $\mathrm{Pos}(u, v, t) = 1 + u^2 + v^2 + t(t + 1)$. Write

$$F = F_0 + 1, \quad z = z_0 + 1, \quad q = \beta J + 1,$$

where F_0 and z_0 are their three-term nonnegative quadratic parts. Then

$$q - F = \beta J - F_0,$$
$$(K_0 + w)C + (q - F) - z(q - 1) = (K_0 + w)C - F_0 - z_0\beta J.$$

Thus the evaluator never forms the two separate $+1$ gates for F and z. This changes neither the substitution nor the quantified domain. The final unrestricted-integer polynomial uses 167 operations, and its three-equation system uses 162.

4.4 Final ledgers and verification scope

Form	Polynomial degree	Operations	System degree	Operations
12 positive, tightest weight	1069	123	1060	118
12 positive, tight weight	1073	121	1064	116
12 positive, simpler weight	1081	120	1072	115
12 positive, refactored norm	1353	113	1344	108
13 positive, monic merger	319	89	288	81
24 unrestricted integers	1936	167	1920	162

All systems in this table have three equations except the thirteen-variable system, which has four. The degrees are exact. For the degree-1069 and 1073 products, naive propagation through the graph gives 1081; their homogeneous cancellation must be used to obtain the exact degree. The graph metadata records this distinction rather than overwriting the larger syntactic bound.

The package contains three complementary kinds of evidence. First, `code/check_exact_identities.py` expands the small free-polynomial identities over $\mathbb{Z}$ and checks every coefficient. This covers the shared odd-component factors, identity (32), the weight factorizations, and both integer-lift cancellations. Second, independent exact evaluations multiply all eight conjugates in an eight-dimensional algebra and compare that result with the division-free graph. Third, `degree_check.py` evaluates the entire graph on univariate slices over two prime fields. Its convolution uses integer packing with a proved carry bound; it does not use floating point or merely propagate the largest input degree. These checks confirm the degree cancellations and support the separate uniform leading-form proofs.

For the signed polynomial, the independently derived monomial

$$J_x^{1354} h\, i_{\mathrm{int}}^{48} \eta_x^{192} w_{\mathrm{int}}^{162} s_{\mathrm{int}}^{177} z_x^2$$

has total degree 1936 and coefficient

$$2^{257}\, 255^4 (57599 \cdot 73983)^2 \beta^{677}.$$

The coefficient is nonzero for every admissible $\beta > 0$; its modular values agree with two independent full-polynomial slice calculations.

Arithmetic identity checks establish the stated circuit transformations. The root-recovery lemmas, the signed-domain argument, the native compiler, and the calibration of K215 remain mathematical proof dependencies. None of these finite calculations is presented as a new Lean or Rocq theorem, or as a test of whether the consistency polynomial has a solution.

5 Smaller materialized coefficients without prime-power rounding

The new arithmetic forms can use the complete Wang943 coefficient tuple of package 19. A further improvement is possible by combining that package's exact support calculation with the repository's Boolean padding lemma for an arbitrary fixed odd factor [12, 13]. The geometric radix dimensions need not themselves be powers of five. What completeness actually requires is a suitable family of tableau sizes and enough freely selectable dummy bits to tune the actual packed index.

5.1 The literal finite recipe

Retain all 943 directed-signal Wang tiles, their order, the occupied ten-bit edge codes, the four copy tracks, and the native exponent positions of the compact construction. The first tile is Start, the second is the globally impossible forbidden tile, and the remaining tiles are lexicographically ordered. The included 430-action machine table and tile list are checked against one another by an independent enumeration in `code/build_coefficients.py`. The machine table's SHA-256 is

`fdbcef0825886035c3091c8537a69535bc96f9e265f0ae45863543d40ae70090.`

The finite language and its computation semantics are inherited unchanged from package 19; its proof is included as `upstream/wang943_language.md`.

Write $A_c = 1024$. In slot order west, north, east, south, the copy offsets are $1023, 1013, 1003, 983$. The native support is

$$\begin{aligned}\mathcal{E} = {} & [0, 942] \cup [983, 992] \cup [1003, 1032] \\ & \cup \{1063, 1064, 1095, 1096, 2195, 3296\}.\end{aligned}$$

It has 989 elements. If tile i has edge labels $t_{i,0}, \ldots, t_{i,3}$, put $\epsilon_{i,s,j}$ equal to bit j of $t_{i,s} + 512$ and define

$$c_i = 1 + \sum_{s=0}^{3} \sum_{j=0}^{9} \epsilon_{i,s,j} A_c^{1+10s+j} + \sum_{j=41}^{44} A_c^j.$$

The copy at exponent offset$_s + j$ has coefficient $A_c^{1+10s+j}$. The anchors $1095, 1096, 2195, 3296$ have coefficients $A_c^{41}, A_c^{42}, A_c^{43}, A_c^{44}$, respectively, and both dummy coefficients are zero. These rules specify every c_e. The clause mask is

$$\mu = A_c - 2 + \sum_{j=1}^{44} (A_c - 3) A_c^j + 2^{450}(2^{78} - 1).$$

Set, now without five-power rounding,

$$b_c = 529, \qquad L_c = 12031, \qquad d_c = b_c L_c = 6364399, \qquad \rho = 2^{529}, \qquad B = 2^{6364399}.$$

All the following quantities are fixed integers, not unknowns:

$$\begin{aligned} D_C &= \rho^{30} + \rho^{5517} + \rho^{1100} + \rho^{2200} + \sum_{e\in\mathcal{E}} c_e(\rho^{7631-e} + \rho^{8733-e}), \\ D_R &= \rho^{5497}, \\ M_{F,0} &= \mu(\rho^{7631} + \rho^{8733}) + \sum_{j=0}^{9} \rho^{1013+j} + \sum_{j=0}^{9} \rho^{6520+j} + \rho^{2195} + \rho^{3296}, \\ M_C &= B - 1 - \sum_{\substack{e\in\mathcal{E}\\ e\neq 1}} \rho^e - 2\rho^{1063}, \qquad M_{F,\mathrm{native}} = M_{F,0} + 4. \end{aligned}$$

The four ports used by *every* circuit in this delivery are

$$\boxed{\beta = B - 1, \quad K_0 = D_C + BD_R, \quad M_C, \quad M_F = M_{F,\mathrm{native}} + B - 1.} \tag{33}$$

The last port is the shifted high mask. Replacing it by the unshifted mask, or replacing only part of this tuple, defines a different and unjustified instance.

5.2 Why the smaller layout is sufficient

The native clause coefficients are unchanged. With

$$\mathcal{M} = (|\mathcal{E}| + 2)\left(2\sum c_e + 6\right), \qquad T_c = \max(4\mathcal{M} + 8, 2\mu + 4, 16),$$

the exact check gives $\mathrm{bitlength}(T_c) = 529$ and $2^{529} \geq T_c$. Thus the same uniform carry bound holds. The largest unshifted product exponent is $8733 + 3296 = 12029 < 12031$. The earlier exact support proof of target isolation, clean clause bands, and unique anchor alignment uses no five-power property. It remains valid for the new cell length.

In particular, the independent finite checks give

$$\begin{aligned} \{(7631 - e) \bmod L_c : e \in \mathcal{E}\} \cap \{(8733 - e) \bmod L_c : e \in \mathcal{E}\} &= \varnothing, \\ \{(2195 - e) \bmod L_c : e \in \mathcal{E}\} \cap \{(3296 - e) \bmod L_c : e \in \mathcal{E}\} &= \{0\}. \end{aligned}$$

All 69,230 native-basis convolution checks agree with the prescribed overlap and clause contributions. The population and residue conditions are exactly

$$\begin{gathered} 0 < M_C, M_{F,\mathrm{native}} < B - 1, \qquad M_C \equiv 2 \pmod 4, \quad M_{F,\mathrm{native}} \equiv 4 \pmod 8, \\ \mathrm{popcount}(M_C) + \mathrm{popcount}(M_{F,\mathrm{native}}) = d_c. \end{gathered}$$

These calculations verify the finite hypotheses of the inherited soundness proof. That proof applies to arbitrary solutions and does not assume that a recovered tableau size is a prime power.

5.3 A fully explicit completeness controller

Completeness needs a new argument: the old claim that the control multiplier is a unit modulo a power of five is no longer applicable. Factor

$$d_c = 23^2 \cdot 53 \cdot 227, \qquad \lambda = 452453, \qquad k_c = 452452 = \lambda - 1.$$

Trial division verifies that λ is prime. Exact modular exponentiation gives

Modulus	529	53	227	452453
Order of B	11	52	226	452452

Consequently $\mathrm{ord}_{d_c}(B) = 64636$ divides $\lambda - 1$, and $B^{k_c} \equiv 1 \pmod{d_c}$. Furthermore

$$B^{k_c} \bmod \lambda^2 = 155849698116 \neq 1, \qquad v_\lambda(B^{k_c} - 1) = 1.$$

Put $H_0 = 1 + BD_C + B^2(D_R + 1)$. The complete unit check is small:

Prime p	$B \bmod p$	$(B^2 - 1) \bmod p$	$H_0 \bmod p$
23	3	8	8
53	14	36	7
227	20	172	14
452453	167241	264979	406475

Choose the spatial width w_c and temporal height h_c to be sufficiently large powers of λ; independent rectangular padding permits this. Write $N_c = w_c h_c = \lambda^n$, $q = B^{N_c}$, and impose $N_c > d_c\lambda^2$ and $h_c > \lambda$. Since all relevant orders divide $\lambda - 1$,

$$q \equiv B^{w_c} \equiv B \pmod{d_c\lambda}.$$

The exact change in the packed Pell index when a lower dummy bit at exponent $e_* = 1063$ in cell j is switched on is inherited unchanged:

$$\delta R = -\Gamma B^j, \qquad \Gamma = \rho^{e_*}(q^2 - 1)\big[1 + q(D_C + BD_R + B^{w_c})\big].$$

The table proves that Γ is a unit modulo $d_c N_c$.

Lemma 5.1 (Boolean coverage of the actual modulus)**.** *Every residue modulo $d_c N_c$ is the sum of a Boolean subset of*

$$\{B^{2+k_c j} : 0 \le j < T\} \cup \{B^3\}, \qquad T = N_c/\lambda.$$

Every selected cell and its spatial and temporal translations used by the compiler are nonwrapping.

Proof. Let $C_0 = d_c\lambda$, so $d_c N_c = C_0 T$ and $T > C_0$. The valuation just proved and the binomial lifting identity show that B^{k_c} has order T modulo λ^n, and order one modulo d_c. Its first T powers are therefore exactly the T distinct residues $1 + C_0 z$ modulo $C_0 T$.

Normalize the desired residue by B^{-2}. Choose $\varepsilon \in \{0, 1\}$ so that the normalized residue minus εB is nonzero modulo λ. Let a be its least residue modulo C_0. Then $1 \le a < C_0 < T$, $\gcd(a, T) = 1$, and the remaining target has the form $a + C_0 u$. Choose z_0 solving

$$az_0 + \frac{a(a-1)}{2} \equiv u \pmod{T}.$$

Take the a distinct orbit entries with additive coordinates $z_0, z_0+1, \ldots, z_0+a-1$ modulo T, and include B^3 exactly when $\varepsilon = 1$. Their sum is the target. No position is reused.

The largest available position is $2+(\lambda-1)(N_c/\lambda-1) = N_c - N_c/\lambda - \lambda + 3$. Since $h_c > \lambda$ implies $w_c < N_c/\lambda$, its translates by one and by w_c stay below N_c. The extra position 3 is distinct because $k_c > 1$ and satisfies the same conditions. □

For a canonical blank halting tableau, begin with all these dummy bits zero and let its actual packed index be R_0. Apply the lemma to

$$(R_0 - d_c w_c)\Gamma^{-1} \pmod{d_c N_c}.$$

The modified tableau has $R \equiv d_c w_c \pmod{d_c N_c}$, exactly the temporal rotation required by the power $X = 2^R$. The selected dummy bits change no tile, overlap, marker, or carry condition. The unchanged carry estimate preserves the strict slack $2C + F < q/3$ and hence positive transport coordinates. The inherited Pell converse supplies the remaining witnesses.

All dimensions $b_c, L_c, d_c, w_c, h_c, N_c$ are odd. More explicitly, write $u_c = d_c N_c = \log_2 q$. The unchanged low-mask congruences give $R \equiv 3 \pmod 4$, so the canonical choice $w = 2^{R-u_c}$ is a positive integer square. For the canonical even Pell parameter $A_c' = a+2$, the coordinate $c = \psi_{A_c'}(R)$ is odd. Choosing the strong rank $m_c = cR$ makes $i = \psi_{A_c'}(m_c)/c^2$ an odd positive integer, by the binomial expansion in the signed source [10, 11]. The exact population identity still gives $\mathrm{popcount}(R) = 3u_c + 2$, and the half-binomial identity gives $v_2(Y) = 3u_c$; hence the scale $s = Y/q^3$ is odd. This establishes the full adapter needed for the 24-integer construction; the modular checks alone would not establish it.

Theorem 5.2 (Complete coefficient replacement)**.** *The complete tuple* (33) *satisfies the same blank-K215 native compiler interface as package 19, including the signed source's canonical odd and square witness requirements. It may replace that tuple in the compatible arithmetic forms in this report.*

5.4 Actual bytes and the size improvement

The four integers are included as gzip-compressed, minimal unsigned little-endian binary payloads. They are already materialized, not merely specified by a program waiting for K215 to halt.

Port	Bits	Raw bytes	Compressed bytes
β	6,364,399	795,550	806
K_0	9,272,313	1,159,040	15,461
M_C	6,364,399	795,550	1,151
M_F	6,364,400	795,550	901
Total		3,545,690	18,319

Package 19 used $d_c = 9,765,625$ and 5,312,269 raw bytes. The new d_c is smaller by about 34.83%, and the raw payload by about 33.25%. All four fixed ports have smaller numerical magnitudes. Compression size is reported separately; it is not an algebraic coefficient-height measure. The values 529 and 12031 are the least odd choices satisfying the existing carry threshold and existing maximum support bound, respectively. This limited minimality claim fixes the tile language, mask and layout; it is not a minimum over all coefficient encodings.

6 Reproducibility, limitations, and remaining directions

6.1 What is delivered

The archive contains the standalone LaTeX source of this report and its compiled PDF, the editable section sources, twenty literal arithmetic circuits, all four compressed coefficient integers, standard-library Python builders and checkers, and the pinned upstream proof and machine files used in the composition. No repository checkout or network connection is needed to run the numerical and algebraic verification. Upstream files retain their source paths, commit identifiers, checksums, and notices. The inherited formal endpoint is a source record, not a self-contained copy of the whole Lean project.

For a practical entry point, use `signed_separating_saturated24_product.json` for unrestricted integers, or `optimized_tightest_separating12_product.json` for the degree-1069 positive equation. Replacing `product` with `system` selects the corresponding three-equation version. Every circuit explicitly refers to the one delivered coefficient binding. The binding records minimal unsigned little-endian bytes and SHA-256 hashes; `MF` always means the shifted high mask.

From the unpacked archive directory, the standard-library verification and reconstruction entry points are

```
python3 verify.py
python3 reproduce.py
pdflatex -interaction=nonstopmode -halt-on-error con_zfc_synthesis.tex
pdflatex -interaction=nonstopmode -halt-on-error con_zfc_synthesis.tex
```

The first command checks the supplied artifacts. The second regenerates the new coefficient tuple and the four principal construction families, then rebinds and verifies the delivered circuits. The six exact-refactored comparison circuits and the two inherited comparison circuits are included literally and checked against their preserved source packets; they do not depend on a missing external generator. The final two commands rebuild the document. The Python scripts require no third-party Python package. The LaTeX source uses ordinary TeX Live packages.

6.2 The verification boundary

The finite audits check the literal machine-to-tile enumeration, the layout's support and carry inequalities, the modular padding constants, the four materialized payloads, every circuit's well-formedness and live inputs, exact small polynomial identities, and independent evaluations of the norm constructions. Full univariate finite-field calculations use the actual four coefficient residues to check the principal degree claims. The positive tight-weight and signed-variable sections also give uniform nonzero leading forms, so the claimed degrees are not based only on random evaluations or generic syntactic upper bounds.

These checks have specific limits. Finite-field agreement is a diagnostic unless backed by the displayed polynomial identity or degree proof. A successful program run does not prove universal compiler soundness, does not formally verify the root-recovery arguments, and does not establish the truth of consistency. The infinite claims rest on the mathematical proofs and their stated inherited interfaces. This report makes no new end-to-end Lean or Rocq verification claim.

The supplied collection is itself a set of research reports; its intake note explicitly distinguishes claims in the packages from independently re-derived facts [1]. This synthesis follows that distinction: it preserves the exact sources for review and supplies new proofs for the changed constructions, rather than treating comparison metadata as a substitute for the reduction.

6.3 An attempted extra variable reduction that is not claimed

The remaining index quotient h is tempting to merge with the main-root quotient. A single combined carrier, however, supplies a combined divisibility condition. Recovering separately the index congruence modulo E and the main-root congruence modulo H would require further arithmetic information, such as an appropriate coprimality condition. There is no available identity forcing $\gcd(E, H) = 1$:

$$E = XY, \qquad H = 4(E + Y) + 3,$$

and the elementary parameter restrictions do not justify that inference. The required decomposition cannot simply be assumed. Consequently no eleven-positive-variable theorem is asserted here. A valid further reduction would have to preserve both congruences, or prove an additional property of every recovered Pell witness before using it.

There are still concrete directions for improvement. The 943-tile language and its spatial exponent layout were kept fixed, so reducing the local alphabet or redesigning the layout could lower coefficient height further. A different norm elimination might improve the tradeoff between twelve-variable degree and arithmetic length. Fewer integer coordinates would require a replacement for one of the six positivity guards, another valid root merger, or a stronger signed interface. None of these potential improvements is counted in the delivered results.

References

[1] ProveIt. *Con(ZFC) in Diophantine form: wave-3 reconciliation*. Pinned repository source at `ae3007ca4467`. Local copy: `upstream/collection_at_pin.md`.

[2] ProveIt. *Package 18: Three explicit Diophantine forms of ZFC consistency*. Pinned repository source at `ae3007ca4467`.

[3] ProveIt. *Package 19: Twelve positive variables and materialized coefficients*. Pinned repository source at `ae3007ca4467`.

[4] ProveIt. *Package 20: Twelve positive unknowns and materialized coefficients*. Pinned repository source at `ae3007ca4467`.

[5] ProveIt. *Package 21: A twelve-variable equation for Con(ZFC)*. Pinned repository source at `ae3007ca4467`. Local copy: `upstream/norm12_proof.md`.

[6] ProveIt. *Package 22: Monic merger proof*. Pinned repository source at `ae3007ca4467`. Local copy: `upstream/monic_merger_proof.md`.

[7] ProveIt. *Package 23: Separating-root twelve-variable construction*. Pinned repository source at `ae3007ca4467`.

[8] ProveIt. *KEndpoint215: blank halting and ZFC refutation*. Pinned repository source at `ae3007ca4467`. Local copy: `upstream/KEndpoint215.lean`.

[9] ProveIt. *Fixed blank 67-operation polynomial and native compiler interface*. Pinned repository source at `ae3007ca4467`. Local copy: `upstream/fixed_blank_67_proof.md`.

[10] ProveIt. *Signed-domain proof for the normalized Pell interface*. Pinned repository source at `ae3007ca4467`. Local copy: `upstream/signed_domain29.md`. The ordinary and helical compiler proofs are also included as `upstream/ordinary_blank_proof.md` and `upstream/blank_helical_proof.md`.

[11] ProveIt. *Odd-scale 27-integer proof and negative-scale exclusion*. Pinned repository source at `ae3007ca4467`. Local copy: `upstream/odd_scale27.md`.

[12] ProveIt. *Wang943 coefficient language and compact-layout proof*. Pinned repository source at `ae3007ca4467`. Local copy: `upstream/wang943_compact_layout.md`. The separate language proof is included as `upstream/wang943_language.md`.

[13] ProveIt. *Pell-kernel prime padding with an arbitrary fixed odd factor*. Pinned repository source at `ae3007ca4467`. Local copy: `upstream/prime_padding.md`.

[14] ProveIt. *Retained-variable 59-operation system*. Pinned repository source at `ae3007ca4467`. Local copy: `upstream/retained59_proof.md`.